Full results for large-sized datasets

Overview datasets

Dataset	n	d	Type	Source
Survival	110,204	4	Tabular	UCI ML Repository
Diabetes	253,680	22	Tabular	UCI ML Repository
Music	515,344	91	Tabular	Kaggle
Covtype	581,012	55	Tabular	UCI ML Repository
Imagenet8	1,281,166	192	Image	ImageNet Image Database
Imagenet32	1,281,166	3,072	Image	ImageNet Image Database
Census	2,458,284	68	Tabular	UCI ML Repository
Finance	6,362,620	12	Tabular	Kaggle

Metric: Cityblock (L1, Manhattan)

Objective function values

					Deviation from best objective (%)						
Dataset	n	d	Best objective	UB gap (%)	HEX-L	ABM-d1000	ABM-d2000	ABM-d3000	ABM-d4000	ABM-d5000	RND
Survival	110,204	4	271,069.32	—	1.35	0.02	0.00	0.00	0.00	0.00	31.31
Diabetes	253,680	22	3,179,814.96	—	20.06	0.03	0.00	0.00	0.00	0.00	29.13
Music	515,344	91	28,304,179.67	—	16.94	1.25	0.68	0.37	0.16	0.00	17.44
Covtype	581,012	55	9,031,204.35	—	21.02	0.28	0.11	0.05	0.02	0.00	22.10
Imagenet8	1,281,166	192	44,254,178.48	—	26.62	1.25	0.66	0.35	0.15	0.00	26.68
Imagenet32	1,281,166	3,072	770,120,739.88	—	—	1.28	0.68	0.36	0.16	0.00	24.22
Census	2,458,284	68	85,474,467.03	—	26.45	0.21	0.09	0.04	0.02	0.00	26.47
Finance	6,362,620	12	24,770,280.56	—	22.36	0.01	0.00	0.00	0.00	0.00	22.40
Average					—	0.54	0.28	0.15	0.06	0.00	24.97

Runtimes (seconds)

			Runtime (s)						
Dataset	n	d	HEX-L	ABM-d1000	ABM-d2000	ABM-d3000	ABM-d4000	ABM-d5000	RND
Survival	110,204	4	7,203.67	0.51	1.56	3.90	5.45	9.00	0.01
Diabetes	253,680	22	7,241.69	1.74	6.79	16.83	29.31	49.66	0.02
Music	515,344	91	7,519.63	3.50	11.34	22.69	37.41	56.33	0.04
Covtype	581,012	55	7,416.01	4.74	18.31	39.76	75.27	122.31	0.04
Imagenet8	1,281,166	192	8,858.97	9.48	31.95	65.73	112.75	161.52	0.10
Imagenet32	1,281,166	3,072	—	61.27	134.53	223.05	308.88	407.88	0.10
Census	2,458,284	68	8,355.82	25.03	102.06	224.31	409.41	663.67	0.19
Finance	6,362,620	12	7,969.95	67.57	276.17	606.69	1,145.63	1,761.84	0.49
Average			—	21.73	72.84	150.37	265.51	404.03	0.12

Metric: Euclidean (L2)

Objective function values

					Deviation from best objective (%)						
Dataset	n	d	Best objective	UB gap (%)	HEX-L	ABM-d1000	ABM-d2000	ABM-d3000	ABM-d4000	ABM-d5000	RND
Survival	110,204	4	184,106.25	-0.73	3.31	0.02	0.01	0.00	0.00	0.00	25.48
Diabetes	253,680	22	995,618.11	-2.12	12.66	0.52	0.25	0.13	0.05	0.00	18.78
Music	515,344	91	3,743,627.11	-3.77	14.99	1.01	0.54	0.29	0.12	0.00	15.47
Covtype	581,012	55	3,065,968.79	-3.20	11.56	0.31	0.15	0.08	0.03	0.00	12.31
Imagenet8	1,281,166	192	3,660,509.02	—	23.46	1.11	0.59	0.31	0.13	0.00	23.48
Imagenet32	1,281,166	3,072	15,942,857.72	—	—	1.09	0.58	0.31	0.13	0.00	20.61
Census	2,458,284	68	16,300,088.82	—	15.55	0.24	0.11	0.06	0.02	0.00	15.59
Finance	6,362,620	12	13,553,068.41	—	20.01	0.00	0.00	0.00	0.00	0.00	20.04
Average					—	0.54	0.28	0.15	0.06	0.00	18.97

Runtimes (seconds)

			Runtime (s)						
Dataset	n	d	HEX-L	ABM-d1000	ABM-d2000	ABM-d3000	ABM-d4000	ABM-d5000	RND
Survival	110,204	4	7,203.36	0.82	3.08	8.85	12.67	21.39	0.01
Diabetes	253,680	22	7,241.47	1.79	6.56	15.47	26.49	43.36	0.02
Music	515,344	91	7,518.42	3.84	12.81	24.84	42.18	65.96	0.04
Covtype	581,012	55	7,414.50	5.85	22.71	47.45	85.68	133.88	0.04
Imagenet8	1,281,166	192	8,883.62	10.06	34.36	69.57	122.50	179.50	0.10
Imagenet32	1,281,166	3,072	—	58.77	132.63	221.03	312.60	416.64	0.10
Census	2,458,284	68	8,351.24	26.67	108.01	237.42	435.50	700.26	0.20
Finance	6,362,620	12	7,973.17	71.82	279.02	610.52	1,151.63	1,770.96	0.50
Average			—	22.45	74.90	154.39	273.66	416.49	0.13

Metric: Squared Euclidean (L2 squared)

Objective function values

					Deviation from best objective (%)							
Dataset	n	d	Best objective	UB gap (%)	HEX-L	ABM-d1000	ABM-d2000	ABM-d3000	ABM-d4000	ABM-d5000	FAC-5	RND
Survival	110,204	4	663,840.25	-6.63	2.43	0.03	0.01	0.00	0.00	0.00	4.60	33.53
Diabetes	253,680	22	7,979,938.03	-4.38	20.31	0.73	0.33	0.16	0.06	0.00	—	30.05
Music	515,344	91	61,814,684.86	-5.16	23.34	2.02	1.10	0.59	0.25	0.00	—	24.15
Covtype	581,012	55	37,666,117.82	-1.63	14.24	0.41	0.20	0.10	0.04	0.00	—	15.16
Imagenet8	1,281,166	192	23,109,921.67	-5.22	42.13	2.28	1.21	0.65	0.27	0.00	—	42.36
Imagenet32	1,281,166	3,072	428,597,408.33	—	—	2.24	1.19	0.64	0.27	0.00	—	38.01
Census	2,458,284	68	219,878,062.39	—	23.92	0.32	0.14	0.07	0.03	0.00	—	23.98
Finance	6,362,620	12	88,425,514.07	—	13.59	0.00	0.00	0.00	0.00	0.00	—	13.65
Average					—	1.00	0.52	0.28	0.12	0.00	—	27.61

Runtimes (seconds)

			Runtime (s)							
Dataset	n	d	HEX-L	ABM-d1000	ABM-d2000	ABM-d3000	ABM-d4000	ABM-d5000	FAC-5	RND
Survival	110,204	4	7,204.08	0.66	2.23	5.86	8.13	13.94	2.89	0.01
Diabetes	253,680	22	7,241.48	1.69	6.24	14.58	24.85	41.67	—	0.02
Music	515,344	91	7,524.23	3.35	10.92	21.66	34.61	53.22	—	0.04
Covtype	581,012	55	7,417.57	5.49	21.34	43.84	81.94	126.70	—	0.04
Imagenet8	1,281,166	192	8,869.02	8.47	27.97	55.34	95.51	143.84	—	0.10
Imagenet32	1,281,166	3,072	—	58.02	127.72	210.64	292.38	386.26	—	0.10
Census	2,458,284	68	8,371.66	26.55	109.67	241.25	435.05	702.47	—	0.19
Finance	6,362,620	12	7,983.34	70.86	283.36	616.68	1,157.23	1,773.48	—	0.49
Average			—	21.89	73.68	151.23	266.21	405.20	—	0.12